

- Q.25 Explain interrupt & types of interrupts in 8051.
- Q.26 Explain different timer modes of 8051.
- Q.27 Write a note addressing modes of 8051.
- Q.28 Explain PUSH and POP instructions with example.
- Q.29 Differentiate between ADD and ADDC instructions.
- Q.30 What is the difference between Byte-Level and Bit-Level instructions.
- Q.31 Write a short note on interfacing in microcontroller systems.
- Q.32 Explain role of A/D convertor in embedded system.
- Q.33 Explain need for patient data acquisition system in biomedical application.
- Q.34 Explain PIC microcontroller & its advantages.
- Q.35 Explain main features 8051 microcontroller.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain pin diagram of 8051 and explain function of each port.
- Q.37 Explain biomedical application of microcontroller with special reference to patient monitoring system.
- Q.38 Explain LED and LED interfacing with 8051.

No. of Printed Pages : 4
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4th Sem / Medical Electronics Sub : Micro controllers applications in medical Technology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A microcontroller mainly consists of
- Only CPU
 - CPU+RAM+ROM+I/O Ports
 - Only ALU
 - Only Memory
- Q.2 8051 Microcontroller has how many I/O Ports
- 2
 - 3
 - 4
 - 5
- Q.3 The size of internal RAM in 8051 is
- 64 bytes
 - 128 bytes
 - 256 bytes
 - 512 bytes
- Q.4 The program counter in 8051 is
- 4 bit
 - 8 bit
 - 16 bit
 - 32 bit

(20)

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183243

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183243

Q.5 In biomedical application, microcontrollers are commonly used for

- a) Patient data acquisition
- b) Broadcasting
- c) Car racing
- d) None of above

Q.6 External memory is enabled in 8051 using

- a) EAPin b) RST Pin
- c) XTAL Pin d) TXD Pin

Q.7 Which addressing mode uses # symbol

- a) Direct b) Register
- c) Immediate d) Indirect

Q.8 LCD interfacing with 8051 is generally done through

- a) I/O ports b) Timer
- c) Interrupt d) Stack

Q.9 A/D Converter is used for conversion of

- a) Analog to Digital b) Digital to analog
- c) AC to DC d) DC to AC

Q.10 The reset pin of 8051 is active

- a) Low b) High
- c) Both d) None

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Expand following terms:-

Q.11 ADC.

Q.12 DAC.

Q.13 LCD

Q.14 ISR.

Q.15 LED.

Q.16 IC.

Q.17 BCD.

Q.18 Expand EMI.

Q.19 Expand PIC.

Q.20 What is EXD.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain basic block diagram of a microcontroller.

Q.22 Explain difference between microprocessor and microcontroller

Q.23 Write a note function of stack pointer.

Q.24 Write a note on need for ceramic resonator / crystal in 8051.

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